

1 Voting Theory

1.5 Voting Theory Review

Example 1.5.1 Use the Plurality method to determine the winner of this election.

	9	19	11	8
1st	B	A	C	B
2nd	A	B	B	C
3rd	C	C	A	A

Example 1.5.2 With the same preference schedule as in the previous example, who wins under the Pairwise Comparison method?

Example 1.5.3 Determine the winner of the election using the Borda Count method.

	11	7	7	3
1st	A	C	B	A
2nd	C	B	C	B
3rd	B	A	A	C

Example 1.5.4 Use the Plurality with Elimination to determine who wins this election.

	120	50	40	90	60	100
1st	C	B	D	A	A	D
2nd	D	C	A	C	D	B
3rd	B	A	B	B	C	A
4th	A	D	C	D	B	C

Example 1.5.5 Previously, we used the Borda Count method to determine that the winner of this election was C . Did this particular election have a Condorcet Candidate? If so, was the Condorcet Criterion broken?

	11	7	7	3
1st	A	C	B	A
2nd	C	B	C	B
3rd	B	A	A	C

Example 1.5.6 Time to think! Does the Pairwise Comparison Method Satisfy the Majority Criterion? Why or why not?

Example 1.5.7 Last lecture we introduced Arrow’s Impossibility Theorem. To demonstrate just how difficult voting can be, consider the following preference schedule.

	5	5	5
1st	A	C	B
2nd	B	A	C
3rd	C	B	A

Who should win?